Bernoulli-EKF Multi-Target Tracker on SE(3)

A Minimal PMBM Reduction for Manipulation-Scene Object Tracking

1 Problem formulation

1.1 Setting, frames, and per-object state

A calibrated RGB-D camera mounted at a fixed extrinsic $T_{bc} \in \mathrm{SE}(3)$ on a mobile robot observes a tabletop workspace populated by an unknown, time-varying number of rigid objects. We work with three frames throughout:

- the world (map) frame w,
- the robot base frame b, and
- the camera frame c, with T_{bc} a known kinematic constant (read from the URDF; on Fetch-class robots this is a fixed head-to-base transform). Camera-in-base lets us interconvert $T_{co} = T_{bc}^{-1} T_{bo}$ at zero variance.

Two external streams are treated as black boxes:

- **Localization** supplies, at every frame k, the base-in-world Gaussian pose $T_{wb,k} \in \mathrm{SE}(3)$, $\Sigma_{wb,k} \in \mathbb{R}^{6\times 6}$. Crucially, only its *relative* change between consecutive frames, $\Delta T_{wb,k} \triangleq T_{wb,k}\, T_{wb,k-1}^{-1} \in \mathrm{SE}(3)$, enters the recursion as a control input (§3, eq. 10); the absolute $(T_{wb,k}, \Sigma_{wb,k})$ enters *only* at the world-frame output composition step (§10). $\sum_{wb,k}$ *never* appears inside the per-object recursion — this is the key design choice that decouples object-pose tracking from localization uncertainty.
- **Proprioception** supplies the gripper-in-base pose $T_{bg,k} \in \mathrm{SE}(3)$ from joint encoders (deterministic at the Bayesian level). Its relative change $\Delta T_{bg,k} \triangleq T_{bg,k}\, T_{bg,k-1}^{-1}$ is the control input for held / manipulation-set objects.

Per-object state. The tracker maintains, online, a variable-size indexed family of *track parameters*

$$\mathcal{X}_k \;=\; \{X_k^{(i)}\}_{i\in\mathcal{I}_k}, \qquad X_k^{(i)} \;=\; \big(\,\ell^{(i)},\; T_k^{(i)},\; P_k^{(i)},\; r_k^{(i)}\,\big), \tag{1}$$

with four attributes per index: a class label $\ell^{(i)} \in \mathcal{L}$ (fixed at birth), a *base-frame pose* $T_k^{(i)} \in \mathrm{SE}(3)$ (object-in-base, $T_{bo}^{(i)}$ in the long form), its covariance $P_k^{(i)} \in \mathbb{R}^{6\times 6}$ in the $\mathfrak{se}(3)$ tangent at $T_k^{(i)}$, and an existence probability $r_k^{(i)} \in [0,1]$ (the prior-probability that the object referred to by index i actually exists at time k). The index set $\mathcal{I}_k \subseteq \mathbb{N}$ is maintained by the tracker (birth enlarges it, pruning shrinks it).

The physical multi-target state at time k is a *random finite set* $\mathcal{T}_k \subseteq \mathrm{SE}(3)$ of *base-frame* poses, one per object actually present. Its (unknown) cardinality is $n_k = |\mathcal{T}_k|$. The parameter family $\mathcal{X}_k$ is the tracker's summary of the posterior on $\mathcal{T}_k$ *conditioned on the localization stream* $\{T_{wb,k}, \Sigma_{wb,k}\}_k$; the relationship is made precise in §4. World-frame quantities $T_{wo}^{(i)}$ are recovered from the base-frame state by composition with $T_{wb,k}$ at output (§10).

Why base frame. The naïve world-frame alternative (store $T_{wo}^{(i)}$, fold $\Sigma_{wb,k}$ into the measurement noise R_k via $R_{\text{eff}} = R_{\text{icp}} + \Sigma_{wb,k}$) is correct for one frame but *double-counts* $\Sigma_{wb,k}$ across consecutive fusions, because $T_{wb,k}$ and $T_{wb,k+1}$ are nearly the same random variable. That recursion drives Σ_{wo} below Σ_{wb}, which is physically impossible for a static object. Storing in base frame removes $\Sigma_{wb,k}$ from the recursion entirely; it appears only once, in the world-frame composition at output, where it correctly lower-bounds Σ_{wo}.

A per-object shape summary $\mathcal{G}_k^{(i)}$ (TSDF volume plus accumulated masked point cloud, rigidly attached to $T_k^{(i)}$) is maintained as a *by-product* of tracking: fused from depth after the pose is settled and $r_k^{(i)}$ is high, and fed back into the recursion only through a predicted-visibility predicate (§9). Geometric relations among objects and manipulation-role tags are deterministic predicates over $\mathcal{X}_k$, computed after each recursion step and consumed by downstream modules; they do not participate in the Bayesian update.

1.2 Per-frame measurements

Each frame delivers

$$Z_k \;=\; \big(I_k,\; D_k,\; \mathcal{D}_k\big), \qquad \mathcal{D}_k \;=\; \big\{\,(m_{k,l},\; \hat{\ell}_{k,l},\; \hat{s}_{k,l},\; \tau_{k,l})\,\big\}_{l=1}^{M_k}, \tag{2}$$

where I_k and D_k are RGB and depth images, and each entry of $\mathcal{D}_k$ is a binary mask $m_{k,l} \in \{0,1\}^{H\times W}$ with class label $\hat{\ell}_{k,l} \in \mathcal{L}$, calibrated detection score $\hat{s}_{k,l} \in [0,1]$, and a video-tracklet identifier $\tau_{k,l} \in \mathbb{N}$ issued by the segmenter (SAM2), which mostly preserves object identity across frames but is occasionally reset when the segmenter loses track – it enters the assignment cost (26) as a matching-ID bonus. All four fields are produced by the open-vocabulary detector + video-segmenter pipeline. **Detections live in pixel space; they do not carry 3D pose information.**

1.3 The mask-to-pose bottleneck

The feature distinguishing this setting from classical multi-target tracking is that *data association must be resolved before any pose measurement exists.* The segmenter tells us the image-space extent of each object hypothesis; converting that to a 3D observation requires a registration target, and that target is the tracker's own shape model $\mathcal{G}_{k-1}^{(i)}$. Each frame therefore executes

$$\mathcal{D}_k \xrightarrow{\text{back-projection}} \big\{\mathcal{C}_{k,l}\big\} \xrightarrow{\text{association}} \pi^* \xrightarrow{\text{ICP vs } \mathcal{G}_{k-1}^{(i)}} \big\{(\hat{T}_{k,l},\, \hat{R}_{k,l})\big\} \xrightarrow{\text{filter update}} \mathcal{X}_k. \tag{3}$$

Concretely, for each mask $m_{k,l}$:

1. **Back-projection.** The masked pixels of D_k yield a *camera-frame* point cloud $\mathcal{C}_{k,l} \subset \mathbb{R}^3$ via the intrinsics K and the depth values; localization is *not* needed for this step.
2. **Association.** A cost-matrix assignment (§6) decides whether $\mathcal{C}_{k,l}$ belongs to an existing track i, or spawns a new one.
3. **Pose measurement.** For a matched pair, ICP of $\mathcal{C}_{k,l}$ against $\mathcal{G}_{k-1}^{(i)}$ (the latter expressed in the object-local frame and initialised at the prior mean $T_{k|k-1}^{(i)}$ projected to camera space) produces a camera-frame pose estimate $\hat{T}_{k,l}^{co} \in \mathrm{SE}(3)$ and a covariance $\hat{R}_{k,l}^{co} \in \mathbb{R}^{6\times 6}$ derived from ICP fitness / residual. We then promote both to the base frame using the fixed kinematic T_{bc}:

$$\hat{T}_{k,l} \;=\; T_{bc}\,\hat{T}_{k,l}^{co}, \qquad \hat{R}_{k,l} \;=\; \mathrm{Ad}(T_{bc})\,\hat{R}_{k,l}^{co}\,\mathrm{Ad}(T_{bc})^{\top}. \tag{4}$$

$\Sigma_{wb,k}$ is *not* folded into $\hat{R}_{k,l}$ here – it plays no role until output composition (§10). For an unmatched mask we take $\hat{T}^{co}_{k,l}$ from the centroid of $\mathcal{C}_{k,l}$ with identity rotation and inflate $\hat{R}^{co}_{k,l}$ as a birth prior.

Only after (3) runs does the filter consume $(\hat{T}_{k,l}, \hat{R}_{k,l})$ as the measurement of an EKF update (§3). The tracker is therefore a feedback loop in which yesterday's state is what makes today's measurement well-defined.

1.4 Estimation target and pathologies

Conditioned on the localization and proprioception streams, the Bayesian target is the multi-target set posterior over *base-frame* poses

$$f_k(\mathcal{T}) \;\triangleq\; p\Big(\mathcal{T}_k = \mathcal{T} \;\Big|\; Z_{1:k},\, T_{wb,1:k},\, \Sigma_{wb,1:k},\, T_{bg,1:k}\Big). \tag{5}$$

The tracker represents f_k by the family $\mathcal{X}_k$ of (1) under the Multi-Bernoulli assumption of §4. Downstream we report the maximum-likelihood cardinality $\hat{n}_k$, per-track labels $\ell^{(i)}$, pose Gaussians $(T^{(i)}_k, P^{(i)}_k)$, and existence probabilities $r^{(i)}_k$.

The recursion must absorb four failure modes simultaneously:

(i) *missed detections* – an object is present but yields no mask, either because the detector failed (probability $1 - p_d$) or because it is occluded / out-of-frustum (visibility p_v);

(ii) *clutter* – spurious masks not corresponding to any real object, at intensity λ_c;

(iii) *association ambiguity* – which mask belongs to which track; a wrong answer corrupts *both* the pose belief (wrong measurement) *and* the existence belief (spurious confirmation);

(iv) *model mismatch* – pose changes exceed what $P^{(i)}_{k|k-1}$ predicts, e.g. an object displaced by an unmodelled external agent, or ICP converging to a wrong local minimum.

Data association (iii) is one failure mode of four. The construction below treats it as a single pass through the same Mahalanobis-distance machinery that gates (ii) / (iv) and weights (i): a single scalar statistic d^2 drives every decision.

We formalise the recursion as a *Bernoulli-EKF tracker* on SE(3): per-object Gaussian pose beliefs with explicit existence probabilities, updated via a single-hypothesis Hungarian assignment on the d^2 cost matrix, with birth from unmatched measurements and principled pruning from the existence update. This is the small-cardinality, low-clutter, single-hypothesis limit of the Poisson Multi-Bernoulli Mixture recursion [1, 2] – equivalently, the JIPDA reduction on SE(3).

2 SE(3) preliminaries

We treat poses as elements of the matrix Lie group SE(3) and represent small pose perturbations as 6-vectors $\xi = (\rho, \omega) \in \mathbb{R}^6$, collecting a translation $\rho \in \mathbb{R}^3$ and a rotation vector $\omega \in \mathbb{R}^3$. The mapping $\xi \mapsto \xi^\wedge \in \mathfrak{se}(3)$ to the Lie algebra $\mathfrak{se}(3)$ is the standard one [4]; its inverse is denoted $\xi = (\xi^\wedge)^\vee$. We invoke $\xi^\wedge$ only to state two basic operations in closed form, after which all computation happens directly in $\mathbb{R}^6$:

$$\mathrm{Exp} : \mathbb{R}^6 \to \mathrm{SE}(3), \qquad \mathrm{Exp}(\xi) \;\triangleq\; \exp(\xi^\wedge), \tag{6}$$

$$\mathrm{Log} : \mathrm{SE}(3) \to \mathbb{R}^6, \qquad \mathrm{Log}(T) \;\triangleq\; \log(T)^\vee. \tag{7}$$

Exp has a closed form (Rodrigues' formula in the rotation block, with a translation Jacobian) [4]; Log is its inverse. They are the only structural operations on SE(3) we use; group composition is plain matrix multiplication and inversion is the matrix inverse.

A *Gaussian on* SE(3) centred at $T_0 \in \mathrm{SE}(3)$ with covariance $P \in \mathbb{R}^{6\times 6}$ is the distribution of

$$T \;=\; T_0 \,\mathrm{Exp}(\xi), \qquad \xi \sim \mathcal{N}(\mathbf{0}, P), \tag{8}$$

written $T \sim \mathcal{N}(T_0, P)$. The mean lives in SE(3); the uncertainty lives in $\mathbb{R}^6$, attached to the tangent space at T_0.

3 Extended Kalman filter on SE(3) (single target)

Overview and notation

Treat each tracked object in isolation. The pose belief at time k is a Gaussian on SE(3) in the body-tangent parametrisation of §2: the mean is $T_k \in \mathrm{SE}(3)$ (a 4×4 rigid transform of the object frame in the world frame), and the spread around it is the 6×6 covariance $P_k \;=\; \mathrm{cov}\big(\xi_k\big),\;\; \xi_k \sim \mathcal{N}(\mathbf{0}, P_k)$, with $\xi = (\rho, \omega)$ the local twist that perturbs T_k via $T_k\,\mathrm{Exp}(\xi)$. When we write $T_{k|k-1}, P_{k|k-1}$ we mean the *predicted* mean / covariance after the process step but before incorporating the frame-k measurement; $T_{k|k}, P_{k|k}$ are their *posterior* counterparts after the update.

The recursion is the standard Extended Kalman filter on a Lie group, specialised to (a) constant-pose dynamics and (b) right-multiplicative, tangent-Gaussian process / measurement noise. Each frame:

1. **Predict.** Apply the known control input $u_k \in \mathrm{SE}(3)$ (§1.1: $\Delta T_{wb,k}^{-1}$ for world-static objects, $\Delta T_{bg,k}$ for held / manipulation-set objects, $u_k = I$ as the trivial fallback) to the mean by left multiplication, Ad-conjugate the covariance, add Q_k, then cap with the saturation projector Φ. Q_k encodes the residual per-frame dynamics not explained by u_k: tiny for a stationary world-static object (most of Q_k work is done by $u_k = \Delta T_{wb,k}^{-1}$), small slack Q_{manip} for held objects (grip slip / deformation), large during grasp/release transitions when u_k is unreliable. Supplied by the manipulation context (`ekf_se3.process_noise_for_phase`).
2. **Measure.** The frame's mask is back-projected to a *camera-frame* 3-D cloud and registered (ICP, §1.3) against the track's shape summary, yielding $\hat{T}_{k,l}^{co} \in \mathrm{SE}(3)$ and $\hat{R}_{k,l}^{co} \in \mathbb{R}^{6\times 6}$. The fixed kinematic T_{bc} promotes both to the base frame at zero variance: $z_k = T_{bc}\hat{T}_{k,l}^{co}$, $R_k = \mathrm{Ad}(T_{bc})\hat{R}_{k,l}^{co}\,\mathrm{Ad}(T_{bc})^{\top}$ (eq. 4). $\Sigma_{wb,k}$ does *not* enter R_k.
3. **Update.** Linearise both the prior Gaussian and the observation in the tangent at $T_{k|k-1}$. The linearised innovation $\nu_k = \mathrm{Log}\big(T_{k|k-1}^{-1} z_k\big) \in \mathbb{R}^6$ lives in the same 6-vector space as ξ; with $H_k = I_6$ the Kalman gain is $K_k = P_{k|k-1} S_k^{-1}$, $S_k = P_{k|k-1} + R_k$. The mean is corrected on the manifold via right-multiplication by $\mathrm{Exp}(K_k \nu_k)$; the covariance is updated in Joseph form and then Φ-saturated.

Glossary of variables (this section).

T_k, P_k. Posterior *base-frame* object pose mean and tangent covariance at frame k ($T_{bo,k}^{(i)}$ in the long form). $T_k \in \mathrm{SE}(3)$ as a 4×4 matrix; $P_k \in \mathbb{R}^{6\times 6}$ symmetric PD. Maintained recursively (initialised at birth, §8; never measured directly).

$T_{k|k-1}$, $P_{k|k-1}$. Predicted mean / covariance produced by step 1.

u_k. Known control input in SE(3) supplied by external streams:

$$u_k = \begin{cases} \Delta T_{bg,k} = T_{bg,k} T_{bg,k-1}^{-1}, & i \in \text{manipulation set} \\ \Delta T_{wb,k}^{-1} = T_{wb,k-1} T_{wb,k}^{-1}, & \text{otherwise (world-static).} \end{cases}$$

The first form is the gripper change in base (proprioception); the second is the inverse base motion (a world-static object's base-frame pose moves opposite to the base). $u_k = I$ on the first frame (no Δ).

ξ_k, ν_k. $\mathbb{R}^6$ vectors in the body tangent at T_{k-1} and $T_{k|k-1}$ respectively. ξ_k is the *unmodelled* per-frame perturbation (anything not captured by u_k); ν_k is the observed innovation.

Q_k. 6×6 process-noise covariance for the residual after u_k is applied, supplied externally per manipulation phase (`process_noise_for_phase`; not estimated).

z_k, R_k. *Base-frame* measurement and its covariance. $z_k = T_{bc} \hat{T}_{k,l}^{co} \in \mathrm{SE}(3)$ from per-mask ICP against $\mathcal{G}_{k-1}^{(i)}$ promoted to base via T_{bc}; $R_k = \mathrm{Ad}(T_{bc}) \hat{R}_{k,l}^{co} \mathrm{Ad}(T_{bc})^\top$ blends ICP residuals only – $\Sigma_{wb,k}$ is excluded.

T_{bc}. Fixed camera-in-base extrinsic (kinematic constant).

$\Delta T_{wb,k}$, $\Delta T_{bg,k}$. Frame-to-frame deltas of base-in-world (from localization) and gripper-in-base (from proprioception); see §1.1.

S_k, K_k, H_k. Innovation covariance, Kalman gain, observation Jacobian. $H_k = I_6$ to first order in ν_k.

Φ, $P_{\max}$. Covariance-saturation projector (eq. 12) and its design constant $P_{\max} \succ \mathbf{0}$; caps $\operatorname{tr} P$ at $\operatorname{tr} P_{\max}$ so the recursion is self-bounded under arbitrarily long missing-observation streaks.

d_k^2, $\mathcal{L}(z_k \,|\, \cdot)$. Mahalanobis squared distance $\nu_k^\top S_k^{-1} \nu_k$ (eq. 19) and the Gaussian innovation likelihood (eq. 18); both consumed downstream by the Bernoulli existence update (§7) and the Hungarian cost matrix (§6).

The remainder of the section makes the three steps quantitative.

Process and measurement models

Tracking happens in the *robot base frame* throughout. For one rigid object, the process model is a known left-multiplicative control $u_k \in \mathrm{SE}(3)$ followed by $\mathbb{R}^6$ tangent-space process noise applied on the right:

$$T_k \;=\; u_k \, T_{k-1} \, \mathrm{Exp}(\xi_k), \qquad \xi_k \sim \mathcal{N}(\mathbf{0}, Q_k). \tag{9}$$

u_k encodes how a deterministic external action moves the object's *base-frame* pose between frames:

$$u_k \;=\; \begin{cases} T_{bg,k} \, T_{bg,k-1}^{-1}, & i \in \text{manipulation set (held / on / in)}, \\ T_{wb,k}^{-1} \, T_{wb,k-1}, & \text{otherwise (world-static object).} \end{cases} \tag{10}$$

Both expressions are derived below from the constraint that the *world-frame* pose of the object is conserved (static) or identified with the gripper (held). In both cases u_k is the **body-frame** inverse of the base/gripper motion, not the world-frame inverse: $T_{wb,k}^{-1} T_{wb,k-1} \neq T_{wb,k-1} T_{wb,k}^{-1}$ in general. Mistaking the latter for the former produces $\sim$1 m drift on a typical mobile-manipulator trajectory because the conjugate transformation differs by an adjoint rotation – enforced experimentally on the apple_in_the_tray sequence (§14 A.7 caveat).

Both $T_{wb,k}$ and $T_{bg,k}$ are external streams (§1.1); we treat them as point estimates – the implementation feeds in $T_{wb,k}$ as the SLAM mean and $T_{bg,k}$ from joint encoders. Localization uncertainty $\Sigma_{wb,k}$ does *not* enter u_k or Q_k; it appears once at the world-frame output composition (§10).

Justification of the two regimes. *World-static object.* If $T_{wo}^{(i)}$ is constant then $T_{wb,k}\,T_{bo,k}^{(i)} = T_{wb,k-1}\,T_{bo,k-1}^{(i)}$ and so $T_{bo,k}^{(i)} = T_{wb,k}^{-1}\,T_{wb,k-1}\,T_{bo,k-1}^{(i)} = u_k\,T_{bo,k-1}^{(i)}$ with $u_k = T_{wb,k}^{-1}T_{wb,k-1}$. *Held object.* If the object is rigidly attached to the gripper ($T_{go}^{(i)}$ =const), then $T_{bo,k}^{(i)} = T_{bg,k}T_{go}^{(i)} = T_{bg,k}T_{bg,k-1}^{-1}T_{bo,k-1}^{(i)} = u_k\,T_{bo,k-1}^{(i)}$ with $u_k = T_{bg,k}T_{bg,k-1}^{-1}$ – the gripper's relative motion is naturally a left-form transform here because T_{bo} and T_{bg} share the base frame on the left of the composition. Setting $u_k = I$ recovers the standard constant-pose model.

Q_k encodes the residual per-frame dynamics not explained by u_k: small slack for held objects (grip slip), tiny drift for static objects (SLAM noise leaks already captured by u_k), large during grasp/release transitions when the regime is switching and u_k is unreliable. Supplied externally by the manipulation context.

The measurement is an ICP-derived noisy pose lifted to the base frame (cf. §1.3), modelled the same way:

$$z_k \;=\; T_k\,\mathrm{Exp}(\eta_k), \qquad \eta_k \sim \mathcal{N}(\mathbf{0}, R_k), \tag{11}$$

where $z_k = T_{bc}\hat{T}_{k,l}^{co}$ and $R_k = \mathrm{Ad}(T_{bc})\hat{R}_{k,l}^{co}\,\mathrm{Ad}(T_{bc})^\top$ (eq. 4). R_k absorbs only ICP uncertainty; $\Sigma_{wb,k}$ is excluded by construction.

Predict. Eq. (9) composes a known SE(3) left-multiplication u_k with right-multiplicative tangent noise. Writing the prior as $T_{k-1|k-1}\,\mathrm{Exp}(\xi_{k-1|k-1})$ with $\xi_{k-1|k-1} \sim \mathcal{N}(\mathbf{0}, P_{k-1|k-1})$, the predicted pose is

$$\begin{aligned} T_k &= u_k\,T_{k-1|k-1}\;\mathrm{Exp}(\xi_{k-1|k-1})\;\mathrm{Exp}(\xi_k) \\ &= \big[u_k\,T_{k-1|k-1}\big]\;\mathrm{Exp}\big(\xi_{k-1|k-1} + \xi_k + \tfrac{1}{2}[\xi_{k-1|k-1}, \xi_k] + \cdots\big) \end{aligned}$$

by Baker–Campbell–Hausdorff. Dropping second-order and higher brackets (the standard EKF-on-Lie-group linearisation) yields $\xi_{k|k-1} \approx \xi_{k-1|k-1} + \xi_k \sim \mathcal{N}(\mathbf{0}, P_{k-1|k-1} + Q_k)$ in the tangent at the *shifted* mean $u_kT_{k-1|k-1}$. Because u_k acts on the left of the original mean, the original tangent vectors transport to the new mean by the adjoint $\mathrm{Ad}(u_k)$ [4], so the prior covariance reaches the new tangent as $\mathrm{Ad}(u_k)\,P_{k-1|k-1}\;\mathrm{Ad}(u_k)^\top$. Applying the *covariance-saturation projector*

$$\Phi : \mathbb{R}^{6\times 6} \to \mathbb{R}^{6\times 6}, \qquad \Phi(P) \;\triangleq\; \min\big(1,\, \mathrm{tr}\,P_{\max}/\,\mathrm{tr}\,P\big)\,P, \tag{12}$$

for a chosen design constant $P_{\max} \succ \mathbf{0}$ (a static uninformative prior, $\mathrm{tr}\,P_{\max} > 0$), the predict step is

$$T_{k|k-1} = u_k\,T_{k-1|k-1}, \tag{13}$$

$$P_{k|k-1} = \Phi\big(\mathrm{Ad}(u_k)\,P_{k-1|k-1}\;\mathrm{Ad}(u_k)^\top + Q_k\big). \tag{14}$$

When $u_k = I$ this collapses to the constant-pose form $T_{k|k-1} = T_{k-1|k-1}$, $P_{k|k-1} = \Phi(P_{k-1|k-1} + Q_k)$. Φ preserves symmetry and positive-definiteness (scaling by a positive scalar) and caps the trace at $\mathrm{tr}\,P_{\max}$: the recursion is self-bounded under arbitrarily long missing-observation streaks. Without it P grows linearly in the number of missed frames.

Update. The innovation, residual covariance, and Kalman gain are

$$\nu_k \;=\; \mathrm{Log}\big(T_{k|k-1}^{-1}\,z_k\big) \;\in\; \mathbb{R}^6, \qquad S_k \;=\; P_{k|k-1} + R_k, \qquad K_k \;=\; P_{k|k-1}\,S_k^{-1}. \tag{15}$$

The measurement Jacobian is $H_k = I_6$ to first order in ν_k: both z and T live in SE(3), and the Log is evaluated at the prior mean, so the linearisation is exact when $\nu_k = \mathbf{0}$ and degrades smoothly

with $\|\nu_k\|$. The state correction is applied in the tangent and folded back via right-multiplication by Exp; covariance is updated in Joseph form for numerical stability:

$$T_{k|k} = T_{k|k-1}\,\mathrm{Exp}\big(K_k\,\nu_k\big), \tag{16}$$

$$P_{k|k} = \Phi\Big(\,(I_6 - K_k)P_{k|k-1}(I_6 - K_k)^\top + K_k R_k K_k^\top\Big). \tag{17}$$

The same projector Φ is applied after the update so a high-innovation observation cannot inflate $P_{k|k}$ above $P_{\max}$ either.

Innovation likelihood and Mahalanobis statistic. By (11) and the linearisation of Log at $T_{k|k-1}^{-1}T_k$, the innovation satisfies $\nu_k \sim \mathcal{N}(\mathbf{0}, S_k)$ under the null hypothesis $H_0^{(k)}$: "z_k was generated by the track whose prior is $\mathcal{N}(T_{k|k-1}, P_{k|k-1})$." The corresponding Gaussian density on $\nu_k \in \mathbb{R}^6$, which we use as the measurement likelihood for the Bernoulli update below, is

$$\mathcal{L}(z_k \,\big|\, T_{k|k-1}, P_{k|k-1}) \;\triangleq\; \mathcal{N}\big(\nu_k;\, \mathbf{0},\, S_k\big) \;=\; (2\pi)^{-3}|S_k|^{-1/2}\exp\big(-\tfrac{1}{2}d_k^2\big), \tag{18}$$

$$d_k^2 \;\triangleq\; \nu_k^\top S_k^{-1}\nu_k \;\sim\; \chi_6^2 \text{ under } H_0^{(k)}. \tag{19}$$

(18) is exact at $\nu_k = \mathbf{0}$ (where the Jacobian of $z \mapsto \mathrm{Log}(T_{k|k-1}^{-1}z)$ is I_6) and first-order accurate elsewhere; the tangent density agrees with the SE(3) density up to this Jacobian.

Robust update. Define an inner gate $G_{\text{in}} = \chi_6^2(0.95) \approx 12.59$ and an outer gate $G_{\text{out}} = \chi_6^2(0.9997) \approx 25$, and the re-weighting factor

$$w_k \;=\; \begin{cases} 1, & d_k^2 \le G_{\text{in}}, \\ \sqrt{G_{\text{in}}/d_k^2}, & G_{\text{in}} < d_k^2 \le G_{\text{out}}, \\ 0, & d_k^2 > G_{\text{out}}. \end{cases} \tag{20}$$

The update (15)–(17) is then modified as follows:

- If $w_k = 0$ (outer-gate reject): skip the update entirely, setting $T_{k|k} = T_{k|k-1}$, $P_{k|k} = P_{k|k-1}$; z_k is routed to the unassigned-track branch of the Bernoulli update (§7).
- Otherwise ($w_k > 0$): substitute $R_k \leftarrow R_k/w_k$ in (15)–(17). Note that K_k, S_k are recomputed with the inflated R_k/w_k; ν_k is unchanged.

(20) is Huber's redescending M-estimator on the Mahalanobis residual, applied to the re-weighted EKF.

4 Per-object Bernoulli state

To each index $i \in \mathcal{I}_k$ we associate a *Bernoulli component* with parameters $B^{(i)} = (r^{(i)}, \ell^{(i)}, T^{(i)}, P^{(i)})$, representing the single-target set posterior over a per-index random finite set $\mathcal{T}^{(i)} \subseteq \mathrm{SE}(3)$ with set density

$$f^{(i)}(\mathcal{T}) \;=\; \begin{cases} 1 - r^{(i)}, & \mathcal{T} = \varnothing, \\ r^{(i)}\,\mathcal{N}\big(T;\, T^{(i)}, P^{(i)}\big), & \mathcal{T} = \{T\},\ T \in \mathrm{SE}(3), \\ 0, & |\mathcal{T}| \ge 2. \end{cases} \tag{21}$$

Here $\mathcal{N}(\cdot; T^{(i)}, P^{(i)})$ is the Gaussian density on SE(3) of (8). In words: with probability $1 - r^{(i)}$ the i-th track does not exist (empty set); with probability $r^{(i)}$ it exists at exactly one pose $T \in \mathrm{SE}(3)$

distributed according to $\mathcal{N}(\cdot; T^{(i)}, P^{(i)})$. Classical Bernoulli-RFS normalisation $\int f^{(i)}(\mathcal{T})\,\delta\mathcal{T} = 1$ holds by construction (FISST set integral; see [1]). The pair $(T^{(i)}, P^{(i)})$ is the EKF state of §3; $r^{(i)}$ is the new variable PMBM contributes; $\ell^{(i)}$ is fixed at birth and does not enter the RFS marginalisation.

Multi-Bernoulli approximation. We assume the per-index Bernoulli sets $\mathcal{T}^{(1)}, \ldots, \mathcal{T}^{(n)}$ are statistically independent. The physical pose RFS of §1 is then

$$\mathcal{T}_k \;=\; \bigcup_{i \in \mathcal{I}_k} \mathcal{T}_k^{(i)}, \tag{22}$$

and its multi-target set density factorises as the convolution (FISST union) of the per-index Bernoulli densities. Equivalently, for any measurable subset $\mathcal{T} \subseteq \mathrm{SE}(3)$,

$$f_k(\mathcal{T}) \;=\; \sum_{\substack{\{\mathcal{T}^{(i)}\}_{i \in \mathcal{I}_k}: \\ \bigcup_i \mathcal{T}^{(i)} = \mathcal{T}}} \prod_{i \in \mathcal{I}_k} f^{(i)}\big(\mathcal{T}^{(i)}\big), \tag{23}$$

which is the Multi-Bernoulli approximation to (5). Independence is the only approximation introduced relative to a full PMBM mixture; the class labels $\{\ell^{(i)}\}$ ride along as track-identity metadata.

5 Bernoulli prediction

For each $i \in \mathcal{I}_{k-1}$, given a constant survival probability $p_s \in (0, 1]$, prediction is the standard PMBM step (eq. (43) in [2], simplified to constant p_s):

$$\begin{aligned} &(T_{k|k-1}^{(i)}, P_{k|k-1}^{(i)}) \quad \text{via EKF predict (13)–(14) with } u_k^{(i)},\, Q_k^{(i)}, \\ &\qquad r_{k|k-1}^{(i)} = p_s\, r_{k-1|k-1}^{(i)}. \end{aligned} \tag{24}$$

p_s is the (observer-independent) physical survival probability of the object between consecutive frames. For static manipulation scenes, where objects neither move nor vanish on their own, the natural value is $p_s = 1$, which collapses (24) to identity on r. Setting $p_s < 1$ would model spontaneous physical disappearance (rare in our domain). Crucially, p_s is *not* visibility-dependent: occlusion-induced "invisibility" does not affect physical survival; it only affects the observer's ability to detect, which enters the recursion through $\tilde{p}_d$ in the missed update (§7). $u_k^{(i)}$ is the per-track control input (eq. 10) that depends on whether i is in the manipulation set; $Q_k^{(i)}$ is the residual process noise supplied by the manipulation context.

Object birth is folded into data association (§8); we do not maintain a continuous Poisson birth intensity.

6 Data association

For every track–detection pair $(i, l) \in \mathcal{I}_{k-1} \times \{1, \ldots, M_k\}$ compute

$$\nu_{il} = \mathrm{Log}\big(T_{k|k-1}^{(i)\,-1} \hat{T}_{k,l}\big), \quad S_{il} = P_{k|k-1}^{(i)} + \hat{R}_{k,l}, \quad d_{il}^2 = \nu_{il}^\top S_{il}^{-1} \nu_{il}, \tag{25}$$

i.e. (15)–(19) with $(z_k, R_k) \leftarrow (\hat{T}_{k,l}, \hat{R}_{k,l})$ and $(T_{k|k-1}, P_{k|k-1}) \leftarrow (T^{(i)}_{k|k-1}, P^{(i)}_{k|k-1})$. The assignment cost matrix $C \in (\mathbb{R} \cup \{+\infty\})^{|\mathcal{I}_{k-1}| \times M_k}$ combines the Mahalanobis distance with a class-label feasibility gate, a χ^2_6 outer gate, and – when the segmenter exposes a tracklet identifier $\tau_{k,l}$ – an additive bonus on matching IDs:

$$C_{il} \;=\; \begin{cases} d^2_{il} \;-\; \alpha\, \mathbb{1}\big[\tau_{k,l} = \tau^{(i)}_{k-1}\big], & \hat{\ell}_{k,l} = \ell^{(i)} \text{ and } d^2_{il} \le G_{\text{out}}, \\ +\infty, & \text{otherwise.} \end{cases} \tag{26}$$

Here $\tau^{(i)}_{k-1} \in \mathbb{N} \cup \{\varnothing\}$ is the tracklet ID with which track i was last matched ($\varnothing$ if never matched, in which case the indicator is 0), and $\alpha \ge 0$ is a tunable bonus in χ^2_6 units ($\alpha = 0$ drops the ID term; the paper default $\alpha = 4.4$ breaks ties between two same-class tracks whose d^2 differ by a few units without overriding a clear geometric mismatch).

The MAP assignment is

$$\pi^* \;=\; \arg\min_{\pi} \sum_{i \in \mathcal{I}_{k-1}\,:\,\pi(i) > 0} C_{i,\pi(i)}, \qquad \text{s.t. } \pi(i) = \pi(j) > 0 \Rightarrow i = j, \tag{27}$$

with $\pi : \mathcal{I}_{k-1} \to \{0, 1, \ldots, M_k\}$ and $\pi(i) = 0$ encoding "track i unassigned". (27) is solved with scipy's `linear_sum_assignment` (Jonker–Volgenant); any cost-$+\infty$ entry forces $\pi^*(i) = 0$. After solving, set $\tau^{(i)}_k \leftarrow \tau_{k,l}$ on matched pairs and leave $\tau^{(i)}_k = \tau^{(i)}_{k-1}$ on misses (so a transient tracklet drop does not erase the memory); birth initialises $\tau^{(i_{\text{new}})}_k = \tau_{k,l}$. Measurements with no positive-assignment track are available for birth (§8). (27) is the MAP collapse of the PMBM mixture over global associations (JIPDA reduction).

Note on circularity: d^2_{il} depends on $(\hat{T}_{k,l}, \hat{R}_{k,l})$, which is itself produced by ICP against $\mathcal{G}^{(i)}_{k-1}$. The orchestrator breaks the loop by running ICP once per track on a coarsely-gated candidate set (masks whose world-frame centroid is within a fixed window of $T^{(i)}_{k|k-1}$); (26) then refines the selection using the resulting pose measurement. Masks failing the coarse gate for all tracks go to birth with a centroid-initialised $\hat{T}_{k,l}$.

7 Bernoulli measurement update

The updates below are direct Bayes applications conditioned on the MAP association π^*. Let $E^{(i)}$ denote the event "track i exists," with prior $\Pr[E^{(i)}] = r^{(i)}_{k|k-1}$.

Track i associated with $z_{k,l}$ ($\pi^*(i) = l$). Given the MAP assignment, $z_{k,l}$ is drawn either from the track's Gaussian-on-SE(3) (if $E^{(i)}$, detected with prob. p_d) or from clutter (if $\neg E^{(i)}$, density λ_c). By Bayes,

$$\begin{aligned} \Pr[E^{(i)} \,\big|\, z_{k,l} \sim \text{track}\, i] &= \frac{\Pr[z_{k,l} \,\big|\, E^{(i)}]\, \Pr[E^{(i)}]}{\Pr[z_{k,l} \,\big|\, E^{(i)}]\, \Pr[E^{(i)}] + \Pr[z_{k,l} \,\big|\, \neg E^{(i)}]\, \Pr[\neg E^{(i)}]} \\ &= \frac{p_d\, \mathcal{L}_{il}\, r^{(i)}_{k|k-1}}{p_d\, \mathcal{L}_{il}\, r^{(i)}_{k|k-1} + \lambda_c \left(1 - r^{(i)}_{k|k-1}\right)}, \end{aligned} \tag{28}$$

where $\mathcal{L}_{il}$ is (18) evaluated at (i, l). Hence

$$\boxed{r^{(i)}_{k|k} \;=\; \frac{p_d\, \mathcal{L}_{il}\, r^{(i)}_{k|k-1}}{p_d\, \mathcal{L}_{il}\, r^{(i)}_{k|k-1} + \lambda_c \left(1 - r^{(i)}_{k|k-1}\right)}.} \tag{29}$$

The conditional state posterior $p(T^{(i)} \mid E^{(i)}, z_{k,l})$ is the EKF update of (16)–(17) with measurement $(z_k, R_k) = (\hat{T}_{k,l}, \hat{R}_{k,l}/w_{il})$ and Huber weight w_{il} from (20); if $w_{il} = 0$ the pair is rerouted to the unassigned branch below.

Track i unassigned ($\pi^*(i) = 0$). Define the *state-dependent* detection probability $\tilde{p}_d^{(i)} = p_d \cdot p_v^{(i)}$, where $p_v^{(i)}$ is the geometric visibility predicate of §9 evaluated at $T_{k|k-1}^{(i)}$. Under the standard PMBM measurement model this is a state-dependent $p_d(\cdot)$ in PMBM's notation (eq. (2) in [2]); the visibility just makes the dependence explicit. Absence of a track-i measurement is consistent with existence only through the missed-detection mechanism (probability $1 - \tilde{p}_d^{(i)}$); non-existence yields the event trivially. By Bayes on the event $E^{(i)} =$ "track i exists",

$$\boxed{r_{k|k}^{(i)} \;=\; \frac{(1-\tilde{p}_d^{(i)})\, r_{k|k-1}^{(i)}}{(1-\tilde{p}_d^{(i)})\, r_{k|k-1}^{(i)} + \big(1 - r_{k|k-1}^{(i)}\big)},} \tag{30}$$

which is the standard Bernoulli-misdetection update with state-dependent p_d (cf. García-Fernández et al. eq. (31)). The state $(T^{(i)}, P^{(i)})$ inherits from the prediction since no new information arrives: $T_{k|k}^{(i)} = T_{k|k-1}^{(i)}$ and $P_{k|k}^{(i)} = P_{k|k-1}^{(i)}$.

When $p_v^{(i)} \to 0$ (geometric occlusion or out-of-FOV) we have $\tilde{p}_d^{(i)} \to 0$, yielding $r_{k|k}^{(i)} \to r_{k|k-1}^{(i)}$ in (30): *no penalty for not seeing the unseen.* Long-term existence loss for genuinely missing-but-expected tracks comes from accumulating (30) over frames in which $p_v^{(i)}$ is high.

8 Birth: unmatched measurements

Each measurement index l with $\pi^*(i) \neq l$ for every $i \in \mathcal{I}_{k-1}$ is a *candidate birth.* Under the simplest birth model, each mask either corresponds to a newly-appearing object (prior intensity λ_b, measured in units of expected births per frame) or to clutter (intensity λ_c). Treat the detector score $\hat{s}_{k,l} \in [0, 1]$ as a calibrated likelihood ratio between the two: the unnormalised likelihood of the mask under the "new object" hypothesis is proportional to $\hat{s}_{k,l}$, and under the "clutter" hypothesis proportional to 1 (independent of the score). Bayes gives the posterior existence probability of the new track,

$$r^{(i_{\text{new}})} \;=\; \frac{\lambda_b\, \hat{s}_{k,l}}{\lambda_b\, \hat{s}_{k,l} + \lambda_c}, \tag{31}$$

which reduces to $\hat{s}_{k,l}$ when $\lambda_b = \lambda_c$ and smoothly interpolates between a pure-score confidence ($\lambda_b \gg \lambda_c$) and a clutter-dominated fallback ($\lambda_c \gg \lambda_b$). The other parameters of the fresh Bernoulli are initialised from the measurement itself:

$$\ell^{(i_{\text{new}})} = \hat{\ell}_{k,l}, \quad T^{(i_{\text{new}})} = \hat{T}_{k,l}, \quad P^{(i_{\text{new}})} = \hat{R}_{k,l}. \tag{32}$$

Since there is no pose prior on a freshly-born object, the posterior covariance collapses to the measurement noise [3]: this is the Bayesian one-step update of an improper uniform prior by a Gaussian likelihood. The new index i_{new} is added to $\mathcal{I}_k$; λ_b is a calibration constant per environment.

A track is *reported* to downstream consumers (planning, shape fusion, scene-graph computation) iff $r_{k|k}^{(i)} \geq r_{\text{conf}}$ (e.g. 0.5); below this threshold the track is *tentative* – still predicted, associated, and updated, but masked from the report $\widehat{\mathcal{X}}_k$.

9 Visibility predicate p_v

We adopt a bounded, deterministic approximation of the true observer-dependent visibility. Let $K \in \mathbb{R}^{3\times 3}$ be the camera intrinsic matrix and $\Omega_{\text{img}} = [0, W] \times [0, H] \subset \mathbb{R}^2$ the image domain. Define the camera projection

$$\pi_K : \mathrm{SE}(3) \to \mathbb{R}^2 \cup \{\infty\}, \qquad \pi_K(T) \;=\; \begin{cases} (Kp)_{1:2}/(Kp)_3, & p = T_{1:3,4},\ (Kp)_3 > 0, \\ \infty, & \text{otherwise}, \end{cases} \tag{33}$$

i.e. the pinhole projection of the object-origin in the camera frame onto the image plane (with depth-positive gate). For tracks $i \neq j$, let $A_k^{(i)} \subseteq \Omega_{\text{img}}$ be the image-space axis-aligned bounding box of track i's shape summary $\mathcal{G}_{k-1}^{(i)}$ projected through $T_{co}^{(i)} = T_{bc}^{-1} T_{k|k-1}^{(i)}$ (the base-frame mean lifted to the camera frame via the fixed kinematic T_{bc}; localization plays no role here), and $\bar{d}_k^{(i)}$ its mean projected depth. Define the occlusion coefficient

$$\rho_{ij,k} \;\triangleq\; \mathbb{1}\big[\bar{d}_k^{(j)} < \bar{d}_k^{(i)}\big]\, \frac{|A_k^{(j)} \cap A_k^{(i)}|}{|A_k^{(i)}|} \;\in\; [0, 1], \tag{34}$$

which is the fraction of i's projected footprint that is both covered by j in the image and *in front of* i along the viewing ray. Distant or non-overlapping j contribute $\rho_{ij,k} = 0$. Then

$$p_v^{(i)} \;=\; \mathbb{1}\big[\pi_K\big(T_{bc}^{-1}\, T_{k|k-1}^{(i)}\big) \in \Omega_{\text{img}}\big] \;\cdot \prod_{j \in \mathcal{I}_{k-1} \setminus \{i\}} \big(1 - \rho_{ij,k}\big). \tag{35}$$

$p_v^{(i)} = 0$ when the projected centroid leaves the image, and the product shrinks with each in-front occluder. (35) evaluates in $O(|\mathcal{I}_{k-1}|^2)$ on bounding boxes; rendering-based refinement (pixel-exact depth buffers) is a drop-in upgrade that does not change any other equation.

10 Pruning and output

After update, prune any track with $r_{k|k}^{(i)} < r_{\min}$ (e.g. 10^{-3}). A track that survives a long sequence of unobserved-but-expected frames sees its existence collapse via repeated application of (30), and is removed naturally without an ad-hoc M-of-N counter.

The output at frame k is the labelled MAP estimate – in *base frame*, the native frame of the recursion:

$$\widehat{\mathcal{X}}_k = \Big\{ \big(i,\ \ell^{(i)},\ T_{k|k}^{(i)},\ P_{k|k}^{(i)},\ r_{k|k}^{(i)}\big) \;:\; i \in \mathcal{I}_k,\ r_{k|k}^{(i)} \geq r_{\text{conf}} \Big\}, \tag{36}$$

where $T_{k|k}^{(i)} = T_{bo,k|k}^{(i)}$ and $P_{k|k}^{(i)} = P_{bo,k|k}^{(i)}$ are object-in-base.

World-frame composition (the only place $\Sigma_{wb,k}$ enters). Downstream consumers that work in world frame (planning, scene-graph visualisation across episodes) compose with the current localization posterior:

$$T_{wo,k}^{(i)} = T_{wb,k}\, T_{bo,k|k}^{(i)}, \tag{37}$$

$$\Sigma_{wo,k}^{(i)} = \mathrm{Ad}\big((T_{bo,k|k}^{(i)})^{-1}\big)\, \Sigma_{wb,k}\, \mathrm{Ad}\big((T_{bo,k|k}^{(i)})^{-1}\big)^{\top} + P_{bo,k|k}^{(i)}, \tag{38}$$

under the working assumption $\delta_{wb} \perp \delta_{bo}$ (a reasonable approximation given base-frame storage decouples them through the recursion). Eq. (38) lower-bounds Σ_{wo} by the projected Σ_{wb} – exactly the desired physics for a static object observed many times from the same base. World-frame output is for reporting only; it is never re-ingested into the recursion.

The single-frame composition above is the cheap default; it uses only the current $T_{wb,k}$ and is blind to historical SLAM updates. A more accurate world-frame estimate that absorbs the entire pose history (loop closures included) is the chain smoother of §11.

11 Per-track observation chain (canonical world-frame state)

The Bernoulli-EKF of §3 is a one-step Markov filter: $T^{(i)}_{k|k}$ depends on the previous posterior and the current measurement only. That is the right tradeoff for the per-frame association / existence loop, but it has a structural limitation: the control input u_k uses only the SLAM mean $T_{wb,k}$ (and its single predecessor), so any retroactive change to the localization trajectory – a loop closure, a bundle-adjustment refinement, an absolute-pose correction from a fiducial – cannot be folded back into the per-track posteriors that have already absorbed the old T_{wb} stream. The accumulated $P^{(i)}_{k|k}$ is wrong but unrecoverable.

Chain definition. For each track i we additionally maintain an append-only *observation chain*

$$\mathcal{H}^{(i)}_k \;=\; \Big\{\big(k_n,\; \hat{T}^{co,(i)}_{k_n,l},\; \hat{R}^{co,(i)}_{k_n,l}\big) \;:\; n = 1, \ldots, N^{(i)}_k\Big\}, \tag{39}$$

where each entry records (a) the frame index k_n at which track i was matched to a detection, (b) the camera-frame ICP pose $\hat{T}^{co,(i)}_{k_n,l}$ produced by Step 1 of §1.3, and (c) the camera-frame ICP covariance $\hat{R}^{co,(i)}_{k_n,l}$. The chain is *purely perception*: it does not store any localization quantity. The fixed kinematic T_{bc} promotes each entry to base frame at zero variance (eq. 4).

Update rules.

- **Birth** (§8): initialise $\mathcal{H}^{(i_{\text{new}})}_k = \{(k, \hat{T}^{co}_{k,l}, \hat{R}^{co}_{k,l})\}$.
- **Match** (§7, matched branch): append $(k, \hat{T}^{co}_{k,l}, \hat{R}^{co}_{k,l})$ to $\mathcal{H}^{(i)}_{k-1}$.
- **Miss** (§7, miss branch): chain unchanged.
- **Prune** (§10): drop the chain.

Memory grows linearly with the matched-frame count of each track. For long-running scenes a sliding window or a marginalize-into-prior operation (replace the oldest M entries with their fused Gaussian) keeps the chain bounded; for typical tabletop episodes $N^{(i)}_k \leq 10^3$ and no compression is needed.

Post-hoc world-frame composition. Given the current localization posterior over the pose history $p\big(\{T_{wb,k}\}_{k=1:K}\big)$ – its Gaussian posterior in the filter case, the full pose-graph posterior when SLAM has loop closures – the world-frame pose of track i is the smoothing estimate

$$\big(\hat{T}^{(i)}_{wo}, \hat{\Sigma}^{(i)}_{wo}\big) \;=\; \text{Smooth}\big(\mathcal{H}^{(i)}_k,\, T_{bc},\, p(\{T_{wb,k}\}_{k=1:K})\big), \tag{40}$$

which we instantiate as a weighted Karcher-mean fusion of the per-entry world-frame poses

$$\hat{T}^{(i)}_{wo,n} = T_{wb,k_n}\, T_{bc}\, \hat{T}^{co,(i)}_{k_n,l}, \tag{41}$$

$$\hat{R}^{(i)}_{wo,n} = \mathrm{Ad}\big(\hat{T}^{(i)\,-1}_{wo,n}\big)\, \Sigma_{wb,k_n}\, \mathrm{Ad}(\cdot)^\top + \mathrm{Ad}(T_{bc})\, \hat{R}^{co,(i)}_{k_n,l}\, \mathrm{Ad}(T_{bc})^\top, \tag{42}$$

with weights $w_n \propto \mathrm{tr}(\hat{R}^{(i)}_{wo,n})^{-1}$ in the simplest information-weighted form. Iterated as a Gauss-Newton step on SE(3) (re-linearise the innovations $\mathrm{Log}\big(\hat{T}^{-1}_{wo,\mathrm{ref}}\hat{T}_{wo,n}\big)$, take the weighted mean correction, repeat until convergence) this is a small per-track batch problem solved in $O(N^{(i)}_k)$ time.

What this buys.

1. **Loop-closure consistency.** When SLAM revises the pose history (a bundle adjustment shifts T_{wb,k_n} for some $k_n \in \mathcal{H}^{(i)}_k$), the chain is unchanged. Eq. (40) immediately reflects the new pose history without re-running the EKF.
2. **Correct Σ_{wb} correlation.** Two observations at the same base pose share the same $\Sigma_{wb,k}$ (they were taken from the same camera). The chain composition naturally preserves that correlation through the per-entry adjoint of eq. (42); the EKF's one-step Markov form cannot.
3. **Compatibility with object-SLAM backends.** The chain is precisely the set of factors a downstream factor-graph smoother (GTSAM, iSAM2) needs: each entry is a binary factor between a base-pose node and an object-landmark node. Migration to a joint pose+object SLAM is a bookkeeping change, not an algorithmic one.

Relation to the Bernoulli-EKF. The two representations co-exist. Per-frame association, Bernoulli existence updates, and birth / prune all read $(T^{(i)}_{k|k}, P^{(i)}_{k|k})$ from the EKF (the *fast tier*); world-frame consumers and SLAM-aware queries read $(\hat{T}^{(i)}_{wo}, \hat{\Sigma}^{(i)}_{wo})$ from the chain (the *canonical world-frame state*). When a SLAM update lands, the chain composition is recomputed and the result can optionally be projected back to base frame ($T^{(i)}_{bo} = T^{-1}_{wb,k}\hat{T}^{(i)}_{wo}$) and *injected* into the EKF as a re-initialisation; this is the periodic reconciliation pattern already used by the slow tier in the existing implementation.

12 Frame-by-frame algorithm

1. **Predict.** For each $i \in \mathcal{I}_{k-1}$: pick $u^{(i)}_k$ per eq. (10) ($\Delta T_{bg,k}$ for manipulation set, $\Delta T^{-1}_{wb,k}$ otherwise); apply state predict via (13)–(14); advance existence via (24).
2. **Measure.** For every mask $m_{k,l}$ back-project to a *camera-frame* point cloud $\mathcal{C}_{k,l}$. For each candidate track i passing a coarse gate, run colored ICP against $\mathcal{G}^{(i)}_{k-1}$ to obtain $(\hat{T}^{co}_{k,l}, \hat{R}^{co}_{k,l})$, then promote to base via (4): $\hat{T}_{k,l} = T_{bc}\hat{T}^{co}_{k,l}$, $\hat{R}_{k,l} = \mathrm{Ad}(T_{bc})\hat{R}^{co}_{k,l}\,\mathrm{Ad}(T_{bc})^\top$. Compute d^2_{il} via (25) and C_{il} via (26).
3. **Associate.** Solve (27) for π^*.
4. **Update each track i.**
   - if $\pi^*(i) = l$ and $w_{il} > 0$: apply (29) and EKF (16)–(17) with R_{il}/w_{il}; set $\tau^{(i)}_k \leftarrow \tau_{k,l}$; *append* $(k, \hat{T}^{co}_{k,l}, \hat{R}^{co}_{k,l})$ to the observation chain $\mathcal{H}^{(i)}_{k-1}$ (§11);

- else: compute $p_v^{(i)}$ via (35) and apply (30) (leaving $\tau_k^{(i)} = \tau_{k-1}^{(i)}$ and $\mathcal{H}^{(i)}$ unchanged).

5. **Birth.** For each unassigned mask: centroid-initialised $\hat{T}$ and inflated $\hat{R}$, then (31), (32), $\tau_k^{(i_{\text{new}})} \leftarrow \tau_{k,l}$, and $\mathcal{H}_k^{(i_{\text{new}})} = \{(k, \hat{T}_{k,l}^{co}, \hat{R}_{k,l}^{co})\}$.

6. **Prune.** Drop i with $r_{k|k}^{(i)} < r_{\min}$.

7. **Emit.** Report base-frame $\widehat{\mathcal{X}}_k$ via (36); world-frame consumers compose with $T_{wb,k}, \Sigma_{wb,k}$ via (37)–(38).

13 Reduction to existing code

- Bernoulli state $(r^{(i)}, \ell^{(i)}, T^{(i)}, P^{(i)}, \tau^{(i)})$ replaces the current per-track entry; two extra scalar fields added to `rbpf_state.RBPFState`: existence r and last-matched SAM2 tracklet ID τ.
- Predict (24), (13)–(14) edits `rbpf_state.predict_objects`; the saturation projector Φ replaces the bucket-switching in `process_noise_for_phase`.
- Cost matrix (26), assignment (27), and per-pair update (29) + Huber-weighted EKF update replace the hard upstream-ID lookup in `orchestrator._fast_tier` with a soft, ID-bonused Mahalanobis assignment: SAM2 IDs still drive identity when consistent, but association degrades gracefully to pure geometry when SAM2 drops an ID.
- (30) replaces the integer `miss_streak`; pruning at $r_{\min}$ replaces the M-of-N deletion test.
- $p_v^{(i)}$ via (35) adds a new `visibility.py` module (∼60 LOC, depends only on the fixed T_{bc} and the per-object shape summaries – no SLAM input).
- Base-frame storage of $T^{(i)}, P^{(i)}$ aligns with the existing `pose_update.gaussian_state.GaussianState` convention; world-frame composition lives in `collapsed_object_world`. The RBPF Bernoulli path is being migrated to base-frame storage to match.

Calibrated parameters. The recursion is parameterised by seven scalars,

$$p_s \in (0,1], \quad p_d \in (0,1], \quad \alpha \geq 0, \quad \lambda_c > 0, \quad \lambda_b > 0, \quad r_{\text{conf}} \in (0,1), \quad r_{\min} \in (0, r_{\text{conf}}),$$

one symmetric positive-definite matrix $P_{\max} \in \mathbb{R}^{6\times 6}$ (the covariance saturation cap of (12)), and two χ_6^2-quantiles $G_{\text{in}} = \chi_6^2(0.95) \approx 12.59$, $G_{\text{out}} = \chi_6^2(0.9997) \approx 25$. Defaults for static manipulation scenes: $p_s = 1$ (rigid objects do not spontaneously disappear), $p_d = 0.9$, $\alpha = 4.4$ (mild tie-break toward SAM2 ID continuity; calibrate on a held-out sequence), λ_c estimated from a few static frames as the average per-frame detection count divided by image area, $\lambda_b = \lambda_c$, $r_{\text{conf}} = 0.5$, $r_{\min} = 10^{-3}$, $P_{\max} = \text{diag}(\sigma_{\text{trans}}^2 \cdot \mathbf{1}_3, \sigma_{\text{rot}}^2 \cdot \mathbf{1}_3)$ with $\sigma_{\text{trans}} = 0.25\,\text{m}$, $\sigma_{\text{rot}} = \pi/4$. Setting $p_s < 1$ models a known per-frame physical-removal hazard (e.g. unstable platforms, transient occluding clutter) and is purely domain-specific; setting $\alpha = 0$ switches off the SAM2 ID prior entirely and recovers the pure-geometry cost (26).

14 Approximations relative to PMBM

The full PMBM posterior of [2] is

$$f_k(\mathcal{T}) = \sum_{Y \uplus W = \mathcal{T}} \underbrace{f^p(Y)}_{\text{PPP, undetected}} \quad \underbrace{f^{mbm}(W)}_{\text{MB mixture, detected}} \quad , \quad f^{mbm}(W) \propto \sum_j \sum_{X_1 \uplus \cdots \uplus X_n = W} \prod_i w_{j,i}\, f_{j,i}(X_i).$$

Our recursion is its small-cardinality, low-clutter, single-hypothesis limit. We list below every place where it deviates and label each deviation as either an *approximation* (lossy collapse of an exact PMBM construct), a *modelling choice* (different but internally Bayesian model), or a *linearisation* (first-order truncation).

A.1 Single-hypothesis collapse via Hungarian assignment *(approximation).* We retain only the MAP global hypothesis j^* from $\sum_j w_j(\cdot)$, found by linear assignment (27). This is the JIPDA reduction of PMBM ([2] Sec. IV-B): conditional on j^* being the true assignment, all subsequent updates are exact Bayesian. Equivalent to PMBM's mixture only when the MAP assignment is unambiguous.

A.2 Marginalised single-MB existence form *(equivalence at marginal level).* PMBM sets $r^+ = 1$ on a successful association and pushes the ambiguity into the hypothesis weight $w_{j,i}$. After collapse to the MAP, those separately-tracked quantities marginalise to our (29): $r^+ = p_d \mathcal{L}\, r / (p_d \mathcal{L}\, r + \lambda_c(1-r))$. The misdetection update (30) is algebraically identical to PMBM's eq. (31) (with state-dependent $\tilde{p}_d^{(i)}$).

A.3 No PPP for never-detected targets *(modelling choice).* We do not maintain the Poisson intensity $\mu(\cdot)$ that PMBM uses to model targets that exist but have not yet produced a measurement. Equivalent to assuming every object in the scene eventually produces at least one detection, which holds for our manipulation domain (objects are deliberately placed by a human or robot and not invisible at first appearance). Loses PMBM's spatial prior over unseen states.

A.4 Categorical birth from detection score *(modelling choice).* We replace PMBM's birth integral $e(z) = p_d \int p(z|y)\mu(y)dy$ with a per-mask Bayes decision over a binary alternative "new object vs. clutter," using the detector score $\hat{s}_{k,l}$ as the likelihood ratio and rate-constants λ_b, λ_c as the prior odds: (31). Internally Bayesian under this model; not spatially adaptive.

A.5 Multi-Bernoulli factorisation across indices *(modelling choice).* We assume the per-index Bernoullis are statistically independent. PMBM also makes this assumption *within* each global hypothesis j; we make it *globally* after the MAP collapse.

A.6 Constant survival probability with $p_s = 1$ default *(modelling choice).* We follow standard PMBM in keeping p_s observer-independent; for static manipulation scenes the natural value is $p_s = 1$, removing predict-step decay entirely. Visibility-dependent "apparent" survival is recovered automatically through the state-dependent $\tilde{p}_d^{(i)}$ in the missed update, which is admissible PMBM.

A.7 EKF on SE(3) with $H = I_6$ *(linearisation).* Process and measurement noise are right-trivialised in the body tangent; the state composition and the innovation use BCH truncated at first order. Exact at $\nu_k = \mathbf{0}$; smoothly degrading with $\|\nu_k\|$. Standard EKF-on-Lie-group practice.

A.8 Covariance saturation Φ *(modelling choice).* The cap on $\operatorname{tr} P$ at $\operatorname{tr} P_{\max}$ in (12) is a pragmatic stabiliser for arbitrarily long missing-observation streaks; it equals an additive Gaussian prior $P_{\max}$ that is not in standard PMBM but is consistent with a maximum-entropy regularisation.

What this buys, what this costs. Approximations A.1 and A.2 are exactly the JIPDA reduction of PMBM, known to be a tight approximation when MAP associations are unambiguous and lossy when crossing tracks induce hypothesis ambiguity. Modelling choices A.3 – A.6 specialise the standard PMBM model to our domain and are reversible: restoring the PPP, Murty K-best mixture, spatially-adaptive birth, or per-class p_s would re-recover full PMBM at proportional computational cost. Linearisations A.7 – A.8 are inherent to the EKF-on-Lie-group and covariance-bounded layer; they do not change the multi-target structure of the recursion.

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